

① 오답노트.md 어제 오답부터 재풀이 ② 아래 신규 문제는 '상한'(시간 부족 시 신규만 이월) ③ 채점: origin 풀이 / /solve-problem

떠올릴 개념

- $M_n = S_n/n$: $E = \mu$, $Var = \sigma^2/n$ · WLLN: $Var \rightarrow 0 \Rightarrow M_n \rightarrow \mu$
- CLT: $Z_n = (S_n - n\mu)/(\sigma\sqrt{n}) \rightarrow N(0,1)$, 분포 무관 · 정규근사·연속성보정(± 0.5)
- MGF·Chebyshev-a.s.수렴은 시험 X (개념만)

문항 목차 (총 8문항)

- [1] L09 · IPSRP 7.3.2 — 점근 CLT
- [2] L09 · IPSRP 7.3.3 — 점근 CLT
- [3] L09 · IPSRP 7.3.4 — 점근 CLT
- [4] L09 · PSP 9.5.1 — 점근 CLT
- [5] L09 · PSP 9.5.2 — 점근 CLT
- [6] L09 · PSP 10.1.1 — 점근 CLT
- [7] 복습 L02 · PSP 2.3.1 — 조건부·Bayes
- [8] 복습 L02 · PSP 2.4.1 — 조건부·Bayes

Problem 2

The number of accidents in a certain city is modeled by a Poisson random variable with an average rate of 10 accidents per day. Suppose that the number of accidents on different days are independent. Use the central limit theorem to find the probability that there will be more than 3800 accidents in a certain year. Assume that there are 365 days in a year.

Problem 3

In a communication system, each codeword consists of 1000 bits. Due to the noise, each bit may be received in error with probability 0.1. It is assumed bit errors occur independently. Since error correcting codes are used in this system, each codeword can be decoded reliably if there are less than or equal to 125 errors in the received codeword, otherwise the decoding fails. Using the CLT, find the probability of decoding failure.

— 풀이 —

Problem 4

50 students live in a dormitory. The parking lot has the capacity for 30 cars. Each student has a car with probability $\frac{1}{2}$, independently from other students. Use the CLT (with continuity correction) to find the probability that there won't be enough parking spaces for all the cars.

9.5.1 ● The waiting time in milliseconds, W , for accessing one record from a computer database is the continuous uniform $(0,10)$ random variable. The read time R

(for moving the information from the disk to main memory) is 3 milliseconds. The random variable X milliseconds is the total access time (waiting time + read time) to get one block of information from the disk. Before performing a certain task, the computer must access 12 different blocks of information from the disk. (Access times for different blocks are independent of one another.) The total access time for all the information is a random variable A milliseconds.

- (a) What is $E[X]$?
- (b) What is $\text{Var}[X]$?
- (c) What is $E[A]$?
- (d) What is σ_A ? time?
- (e) Use the central limit theorem to estimate $P[A > 116\text{ms}]$.
- (f) Use the central limit theorem to estimate $P[A < 86\text{ms}]$.

9.5.2 ● Internet packets can be classified as video (V) or as generic data (D). Based on a lot of observations taken by the Internet service provider, we have the following probability model: $P[V] = 3/4$, $P[D] = 1/4$. Data packets and video packets occur independently of one another. The random variable K_n is the number of video packets in a collection of n packets.

- (a) What is $E[K_{100}]$, the expected number of video packets in a set of 100 packets?
- (b) What is $\sigma_{K_{100}}$?
- (c) Use the central limit theorem to estimate $P[K_{100} \geq 18]$.
- (d) Use the central limit theorem to estimate $P[16 \leq K_{100} \leq 24]$.

10.1.1 ● $X_1, \dots, X_n$ is an iid sequence of exponential random variables, each with expected value 5.

- (a) What is $\text{Var}[M_9(X)]$, the variance of the sample mean based on nine trials?
- (b) What is $P[X_1 > 7]$, the probability that one outcome exceeds 7?
- (c) Use the central limit theorem to estimate $P[M_9(X) > 7]$, the probability that the sample mean of nine trials exceeds 7.

2.3.1● Consider a binary code with 5 bits (0 or 1) in each code word. An example of a code word is 01010. In each code word, a bit is a zero with probability 0.8, independent of any other bit.

- (a) What is the probability of the code word 00111?
- (b) What is the probability that a code word contains exactly three ones?

2.4.1■ A particular operation has six components. Each component has a failure probability q , independent of any other component. A successful operation requires both of the following conditions:

- Components 1, 2, and 3 all work, or component 4 works.
- Component 5 or component 6 works.

Draw a block diagram for this operation similar to those of Figure 2.2 on page 53. Derive a formula for the probability $P[W]$ that the operation is successful.